

Chapter 3

Lists and Collections: Organize Biological Data

Move from one value to many values: samples, genes, counts, and unique bases.

EDOS learning idea

We will use biology and DNA examples because they are concrete, visual, and easy to experiment with. Type the examples yourself, change one thing, run again, and observe what changed.

What you will be able to do

- Create and modify lists.
- Use `len()`, `append()`, `remove()`, `sort()`, and indexing with lists.
- Understand tuples, dictionaries, and sets at a beginner level.
- Choose a collection that matches the biological question.

Classroom rhythm

Read a small idea -> run the example -> change one value -> predict the result -> run again -> explain what happened in your own words.

3.1 Why lists matter

Biology rarely gives you only one sample. A list lets one variable hold many values in order.

```
samples = ['control_1', 'control_2', 'treated_1']
print(samples)
print(len(samples))
print(samples[0])
```

3.2 List methods: add, remove, sort

```
genes = ['TP53', 'BRCA1', 'MYC']
genes.append('EGFR')
genes.remove('MYC')
genes.sort()
print(genes)
```

`append()` adds one item at the end. `remove()` removes a matching value. `sort()` changes the list into sorted order.

3.3 Lists can hold numbers too

```
expression = [12.4, 20.1, 8.2, 15.7]
print('First value:', expression[0])
print('Number of measurements:', len(expression))
print('Highest:', max(expression))
print('Lowest:', min(expression))
```

Built-in functions such as `max()`, `min()`, and `sum()` are very handy with numeric lists.

3.4 Tuples: values that belong together

A tuple looks like a list but uses parentheses. Beginners can think of it as a small fixed record.

```
sample = ('S01', 'E. coli', 37)
print(sample[0])
print(sample[1])
```

You will often see tuples returned by libraries. You do not need to use them everywhere.

3.5 Dictionaries: label each piece of information

A dictionary stores key-value pairs. It feels a bit like a small record with named columns.

```
sample = {
    'id': 'S01',
    'species': 'E. coli',
    'temperature': 37
}
print(sample['species'])
sample['condition'] = 'control'
print(sample)
```

Dictionaries become extremely useful when data has meaningful field names.

3.6 Sets: keep only unique values

```
dna = 'AAGGCTTTCCAA'
unique_bases = set(dna)
print(unique_bases)
```

A set stores unique values and does not promise a fixed order. Sets are useful for questions like: which base types appear at all?

3.7 Choosing a collection

Use a list when order matters and you have many similar items. Use a dictionary when each value needs a label. Use a set when uniqueness matters. Use a tuple for a small fixed group of values.

Try it yourself

These are practice prompts, not a test. If your code breaks, that is useful information. Read the error, change one thing, and try again.

1. Create a list of four gene names and append a fifth.
2. Create a list of nucleotide counts and find `min()`, `max()`, and `sum()`.
3. Create a dictionary for one DNA sample with `id`, `species`, and `sequence`.
4. Use `set()` to find which letters occur in `AAATTTTGGGCCC`.
5. Make a list of DNA sequences and print the length of the second sequence.

Tiny debugging habit

When something fails, check spelling, quotation marks, brackets, colons, and indentation first. Then print intermediate values so you can see what Python sees.

Chapter summary

- Lists organize ordered collections.
- `append()`, `remove()`, `sort()`, and `len()` are core list tools.
- Dictionaries give names to fields.
- Sets keep unique values.
- Choosing the right collection makes code easier to read.

Next step

Open Notebook 03 in EDOS Jupyter and run the matching exercises. Then continue to Chapter 4.

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